

Shubham Marathe

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

Computer Science, B.S.

September 2023 - June 2027

- GPA: 3.83 | 3x Dean's Honors List: Fall 2023, Winter 2024, Spring 2025
- Relevant Coursework: Data Structures and Algorithms, Principles of System Design, OOP, Computer Architecture, Machine Learning, Computer Networks, Linear Algebra, Discrete Mathematics, Calculus
- Organizations: Computer Networking Student Association, Google Student Developer Club, Santa Cruz AI

SKILLS

Languages: Python, C/C++, Java, RISC-V, HTML/CSS

Technologies: Linux, Unix, Bash, Git, Gitlab

Tools: VS Code, Google Suite, Adobe Creative Cloud

PROJECTS

Multi-Threaded HTTP Server | *C Programming, POSIX Threads, Linux, Socket Programming*

- Built a concurrent HTTP/1.1 server in C that supports simultaneous client connections via a thread-pool architecture with a dispatcher and multiple worker threads
- Designed a thread-safe request queue and implemented per-URI reader-writer locks to enable safe parallel GET/PUT operations without race conditions
- Optimized throughput by parallelizing non-conflicting file requests, improving performance under concurrent workloads

Monte Carlo Tree Search for Ultimate Tic Tac Toe | *Python, NumPy, Adversarial AI*

- Used selection, expansion, rollout, and backpropagation with dynamic tree construction and opponent modeling to simulate adversarial turns
- Developed both vanilla and heuristic-enhanced versions, improving decision quality by incorporating domain-specific rollouts
- Conducted controlled experiments varying tree size and rollout depths, running 100 game tournaments via simulations to quantify win rate improvements and convergence trends

EXPERIENCE

Undergraduate Researcher

January 2026 - Present

RANDLab, UCSC

Santa Cruz, CA

- Building an automated cross-region website diffing framework to analyze 100K+ websites per region across 15 global locations, detecting differences in cookies, third-party trackers, security headers, TLS configurations, and privacy policies.
- Implemented browser-based instrumentation to log and compare hundreds of network requests and cookies per site, quantifying regional changes in tracking behavior and feature availability.
- Designed divergence scoring pipelines to process millions of weekly measurements and identify statistically significant cross-region differences in privacy and security posture.

Baskin Robbins

February 2022 - September 2023

Crew Member

San Ramon, CA

- Prepared and served all menu items, including ice cream scoops, sundaes, milkshakes, and cakes
- Handled cash register operations, processed payments, and balanced tills daily
- Maintained sanitary workstations and food safety standards, completing daily cleaning tasks and restocking checklists